Worksheet 1: Decimals – 2nd ESO

1. Theory

Decimals help us describe numbers more precisely. On the FM radio, frequencies are decimal numbers because they lie between whole numbers (e.g., 87.5 MHz).

Decimal place value

Digits after the decimal point represent tenths, hundredths, thousandths, etc., which helps us specify exact frequencies.

Rounding decimals

In radio tuning, rounding affects the station you receive, so understanding how to round decimals is important.

2. Exercises

Rounding

Round these frequencies to one decimal place:

- 91.67 MHz
- 99.24 MHz
- 103.89 MHz

Ordering

Put these frequencies in order from lowest to highest:

- 88.3 MHz
- 88.1 MHz
- 88.9 MHz
- 88.0 MHz

Conversion: MHz to Hz

- 87.9 MHz = ? Hz
- 105.2 MHz = ? Hz

3. Competence-based problems

Problem 1

A radio station broadcasts at 99.7 MHz. What is this frequency in Hz?

Problem 2

If you increase the frequency by 0.1 MHz, what is the new frequency if the original was 98.3 MHz?

Solutions

Rounding:

- 91.67 → 91.7 MHz
- 99.24 → 99.2 MHz
- 103.89 → 103.9 MHz

Ordering:
88.0 < 88.1 < 88.3 < 88.9 MHz

Conversion:

- 87.9 MHz = 87,900,000 Hz
- 105.2 MHz = 105,200,000 Hz

Problems:

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    1. 99.7 MHz = 99,700,000 Hz
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    2. New frequency = 98.3 + 0.1 = 98.4 MHz
